5: Factors and Multiples

Tune In

1. 24 children participated in the Republic Day Parade.
2. We can find this by counting the number of children in each row and then multiplying it with the number of rows. That makes is $4 \times 6 = 24$.
3. 1 row × 4 students = $1 \times 4 = 4$ students
   2 rows × 2 students = $2 \times 4 = 8$ students
   3 rows × 4 students = $3 \times 4 = 12$ students
   6 rows × 4 students = $6 \times 4 = 24$ students
4. Yes

Let's Try! 1

1.

$$\begin{array}{r} 4 \\ 8\,\overline{)\,32} \\ -32 \\ \hline 0 \\ \hline \end{array}$$

As, 8 divides 32 four times and the remainder is 0, so, 8 is a factor of 32.

2.

$$\begin{array}{r} 5 \\ 9\,\overline{)\,45} \\ -45 \\ \hline 0 \\ \hline \end{array}$$

As, remainder = 0 so, 9 is a factor of 45.

3. Every number is a factor of itself so, 8 is a factor of 8.
4. 1 is a factor of all numbers. So, 1 is a factor of 5.
5.

$$\begin{array}{r} 5 \\ 2\,\overline{)\,11} \\ -10 \\ \hline 1 \\ \hline \end{array}$$

As remainder = 1, so 2 is not a factor 11.

Let's Try! 2

1. Using multiplication, let us find factors of 28.
   $1 \times 28 = 28$; factors 1, 28
   $2 \times 14 = 28$; factors 2, 14
   $3 \times 9 = 27$, remainder 1
   $4 \times 7 = 28$, factors 4, 7
   $5 \times 5 = 25$, remainder 3
   $6 \times 4 = 24$, remainder 3
   There is no need to multiply further as 7 is already a factor.
   So, the factors of 28 are 1, 2, 4, 7, 14, 28.

2. Using division for finding factors of 16,
   $16 \div 1 = 16$; factors 1, 16
   $16 \div 2 = 8$; factors 2, 8
   $16 \div 4 = 4$; factor 4
   So, factors of 16 are 1, 2, 4, 8 and 16

Let's Try! 3

1. **Common factors of 15 and 9**
   Factors of 9 are 1, 3, 9
   Factors of 15 are 1, 3, 5, 15
   Common factors are: 1 and 3 **(Ans)**
2. **Common factors of 16 and 8**
   Factors of 16: 1, 2, 4, 8, 16
   Factors of 8: 1, 2, 4, 8.
   Common factors of 8 and 16 are: 1, 2, 4 and 8. (Ans)

Let's Try! 4

1. 16, 44 and 58 to be circled.
2. 15 and 49 to be circled.

Exercise 5A

6.

a. $1 \times 18 = 18 \rightarrow$ The factors are 1 and 18.
$2 \times 9 = 18 \rightarrow$ The factors are 2 and 9.
$3 \times 6 = 18 \rightarrow$ The factors are 3 and 9.
$\therefore$ The factors of 18 are 1, 2, 3, 6, 9, and 18.

b. $1 \times 45 = 45 \rightarrow$ The factors are 1 and 45.
$3 \times 15 = 45 \rightarrow$ The factors are 3 and 15.
$5 \times 9 = 45 \rightarrow$ The factors are 5 and 9.
$\therefore$ The factors of 45 are : 1, 3, 5, 9, 15, and 45.

c. $1 \times 27 = 27 \rightarrow$ The factors are 1 and 27.
$3 \times 9 = 27 \rightarrow$ The factors are 3 and 9.
$\therefore$ The factors of 27 are 1, 3, 9, and 27.

d. $1 \times 50 = 50 \rightarrow$ The factors are 1 and 50.
$2 \times 25 = 50 \rightarrow$ The factors are 2 and 25.
$5 \times 10 = 50 \rightarrow$ The factors are 5 and 10.
$\therefore$ The factors of 50 are : 1, 2, 5, 10, 25, and 50.

e. $1 \times 35 = 35 \rightarrow$ The factors are 1 and 35.
$5 \times 7 = 35 \rightarrow$ The factors are 5 and 7.
$\therefore$ The factors of 35 are 1, 5, 7, and 35.

f. $1 \times 18 = 81 \rightarrow$ The factor are 1 and 81.
$3 \times 27 = 81 \rightarrow$ The factor are 3 and 27. 9
$\times 9 = 81 \rightarrow$ The factor is 9.
$\therefore$ The factors of 81 are 1, 3, 9, 27, and 81.

g. $1 \times 64 = 64 \rightarrow$ The factors are 1 and 64.
$2 \times 32 = 64 \rightarrow$ The factors are 2 and 32.
$4 \times 16 = 64 \rightarrow$ The factors are 4 and 16.
$8 \times 8 = 64 \rightarrow$ The factor is 8.
$\therefore$ The factors of 64 are 1, 2, 4, 8, 16, 32, and 64.

h. $1 \times 72 = 72 \rightarrow$ The factor are 1 and 72.
$2 \times 36 = 72 \rightarrow$ The factor are 2 and 36.
$3 \times 24 = 72 \rightarrow$ The factor are 3 and 24.
$4 \times 18 = 72 \rightarrow$ The factor are 4 and 18.
$6 \times 12 = 72 \rightarrow$ The factor are 6 and 12.
$8 \times 9 = 72 \rightarrow$ The factor are 8 and 9.
$\therefore$ The factors of 72 are 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, and 72.

7.

a. $42 \div 1 = 42 \rightarrow$ The factors are 1 and 42.
$42 \div 2 = 21 \rightarrow$ The factors are 2 and 21.
$42 \div 3 = 14 \rightarrow$ The factors are 3 and 14.
$42 \div 4 = 10$, $(R = 2) \rightarrow$ 4 is not a factor.
$42 \div 5 = 8$, $(R = 2) \rightarrow$ 5 is not a factor.
$42 \div 6 = 7 \rightarrow$ The factors are 6 and 7.
$\therefore$ The factors of 42 are 1, 2, 3, 6, 7, 14, 21, and 42.

b. $56 \div 1 = 56 \rightarrow$ The factors are 1 and 56.
$56 \div 2 = 28 \rightarrow$ The factors are 2 and 28.
$56 \div 4 = 14 \rightarrow$ The factors are 4 and 14.
$56 \div 5 = 11$, $(R = 1) \rightarrow$ 5 is not a factor.
$56 \div 6 = 9$, $(R = 2) \rightarrow$ 6 is not a factor.
$56 \div 7 = 8 \rightarrow$ The factors are 7 and 8.
$\therefore$ The factors of 56 are 1, 2, 4, 7, 8, 14, 28, and 56.

c. $22 \div 1 = 22 \rightarrow$ The factors are 1 and 22.

$22 \div 2 = 11 \rightarrow$ The factors are 2 and 11.

$22 \div 3 = 7$, $(R = 1) \rightarrow 3$ is not a factor.

$22 \div 4 = 5$, $(R = 2) \rightarrow 4$ is not a factor.

$22 \div 5 = 4$, $(R = 2) \rightarrow 5$ is not a factor.

$22 \div 6 = 3$, $(R = 4) \rightarrow 6$ is not a factor.

$22 \div 7 = 3$, $(R = 1) \rightarrow 7$ is not a factor.

$22 \div 8 = 2$, $(R = 6) \rightarrow 8$ is not a factor.

$22 \div 9 = 2$, $(R = 4) \rightarrow 9$ is not a factor.

$22 \div 10 = 2$, $(R = 2) \rightarrow 10$ is not a factor.

Now the next number is 11 which is a factor of 22 so we will not check any further.

$\therefore$ The factors of 22 are 1, 2, 11, and 22.

d. $39 \div 1 = 39 \rightarrow$ The factors are 1 and 39.

$39 \div 2 = 19$, $(R = 1) \rightarrow 2$ is not a factor.

$39 \div 3 = 13 \rightarrow$ The factors are 3 and 13.

$39 \div 4 = 9$, $(R = 3) \rightarrow 4$ is not a factor.

$39 \div 5 = 7$, $(R = 4) \rightarrow 6$ is not a factor.

$39 \div 6 = 6$, $(R = 3) \rightarrow 6$ is not a factor.

$39 \div 7 = 5$, $(R = 4) \rightarrow 7$ is not a factor.

$39 \div 8 = 7$, $(R = 1) \rightarrow 8$ is not a factor.

$39 \div 9 = 4$, $(R = 3) \rightarrow 9$ is not a factor.

$39 \div 10 = 3$, $(R = 9) \rightarrow 10$ is not a factor.

$39 \div 11 = 3$, $(R = 6) \rightarrow 11$ is not a factor.

$39 \div 12 = 3$, $(R = 3) \rightarrow 12$ is not a factor.

Now the next number is 13 which is a factor of 39 so we will not check any further.

$\therefore$ The factors of 39 are 1, 3, 13, and 39.

e. $63 \div 1 = 63 \rightarrow$ The factors are 1 and 63.

$63 \div 2 = 31$, $(R = 1) \rightarrow 2$ is not a factor.

$63 \div 3 = 21 \rightarrow$ The factors are 3 and 21.

$63 \div 4 = 15$, $(R = 3) \rightarrow 4$ is not a factor.

$63 \div 5 = 12$, $(R = 3) \rightarrow 5$ is not a factor.

$63 \div 6 = 10$, $(R = 3) \rightarrow 6$ is not a factor.

$63 \div 7 = 9$, The factors are 7 and 9.

$63 \div 8 = 7$ $(R = 7) \rightarrow 8$ is not a factor.

Now the next number is 9 which is a factor of 63 so we will not check any further.

$\therefore$ The factors of 63 are 1, 3, 7, 9, 21, and 63.

f. $75 \div 1 = 75 \rightarrow$ The factors are 1 and 75.

$75 \div 2 = 34$, $(R = 1) \rightarrow 2$ is not a factor.

$75 \div 3 = 25 \rightarrow$ The factors are 3 and 25.

$75 \div 4 = 18$, $(R = 3) \rightarrow 4$ is not a factor.

$75 \div 5 = 15 \rightarrow$ The factors are 5 and 15.

$75 \div 6 = 12$, $(R = 3) \rightarrow 6$ is not a factor.

$75 \div 7 = 10$, $(R = 5) \rightarrow 7$ is not a factor.

$75 \div 8 = 9$ $(R = 3) \rightarrow 8$ is not a factor.

Similarly 9, 10, 11, 12, 13, and 14 are not the factors of 75.

Now the next number is 15 which is a factor of 75 so we will not check any further.

$\therefore$ The factors of 75 are 1, 3, 5, 15, 25, and 75.

g. $14 \div 1 = 14 \rightarrow$ The factors are 1 and 14.

$14 \div 2 = 7 \rightarrow$ The factors are 2 and 7.

$14 \div 3 = 4 \rightarrow$ $(R = 2) \rightarrow 3$ is not a factor.

$14 \div 4 = 3$, $(R = 2) \rightarrow 4$ is not a factor.

$14 \div 5 = 2$, $(R = 4) \rightarrow 5$ is not a factor.

$14 \div 6 = 2$, $(R = 2) \rightarrow 6$ is not a factor.

Now the next number is 7 which is a factor of 14 so we will not check any further.

$\therefore$ The factors of 14 are 1, 2, 7, and 14.

h. $32 \div 1 = 32 \rightarrow$ The factors are 1 and 32.

$32 \div 2 = 16 \rightarrow$ The factors are 2 and 16.

$32 \div 3 = 10 \rightarrow (R = 2) \rightarrow 3$ is not a factor.

$32 \div 4 = 8 \rightarrow$ The factors are 4 and 8.

$32 \div 5 = 6$, $(R = 2) \rightarrow 5$ is not a factor.

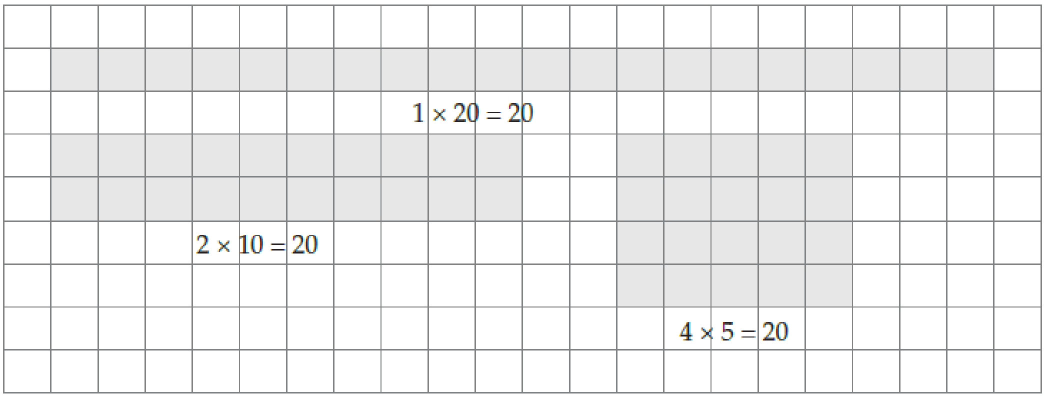
$32 \div 6 = 5$, $(R = 2) \rightarrow 6$ is not a factor.

$32 \div 7 = 40$, $(R = 4) \rightarrow 7$ is not a factor.

Now the next number is 8 which is a factor of 32 so we will not check any further.

$\therefore$ The factors of 32 are 1, 2, 4, 8, 16, and 32

8.



So, the factors of 20 are 1, 2, 4, 5, 10, and 20.

9.

$1 \times 24 = 24 \rightarrow$ The factors are 1 and 24

$2 \times 12 = 24 \rightarrow$ The factors are 2 and 12

$3 \times 8 = 24 \rightarrow$ The factors are 3 and 8

$4 \times 6 = 24 \rightarrow$ The factors are 4 and 6

$\therefore$ The factors of 24 are : 1, 2, 3, 4, 6, 8, 12, and 24.

$1 \times 30 = 30 \rightarrow$ The factors are 1 and 30

$2 \times 15 = 30 \rightarrow$ The factors are 2 and 15

$3 \times 10 = 30 \rightarrow$ The factors are 3 and 10

$5 \times 6 = 30 \rightarrow$ The factors are 5 and 6

$\therefore$ The factors of 30 are : 1, 2, 3, 5, 6, 10, 15, and 30.

Common factors are : 1, 2, 3, 6

Smallest common factor : 1

Greatest common factor : 6

10. Factors of 55 using multiplication are:

$1 \times 55 = 55$; Factors: 1, 55

$5 \times 11 = 55$ Factors: 5, 11

Ans: Factors of 55 are, 1, 5, 11 and 55. It is an odd number. All the factors are odd numbers.

Let's Try! 5

1.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50

2. The first five multiples of 7 are 7, 14, 21, 28 and 35.

Let's Try! 6

1. The first three multiples of 6 are: 6, 12, and 18.
2. The first three multiples of 34 are: 34, 68 and 102.
3. The first three multiples of 100 are: 100, 200 and 300.

Let's Try! 7

1.

×	**1**	**2**	**3**	**4**
2	**2**	**4**	**6**	**8**
6	**6**	**12**	**18**	**24**

The first common multiple of 2 and 6 is 6.

2.

×	1	2	3	4	5	6	7	8	9	10	11	12	13	14
5	5	10	15	20	25	30	**35**	40	45	50	55	60	65	**70**
7	7	14	21	28	**35**	42	49	56	63	**70**	77	84	91	98

The first two common multiples of 5 and 7 are 35 and 70.

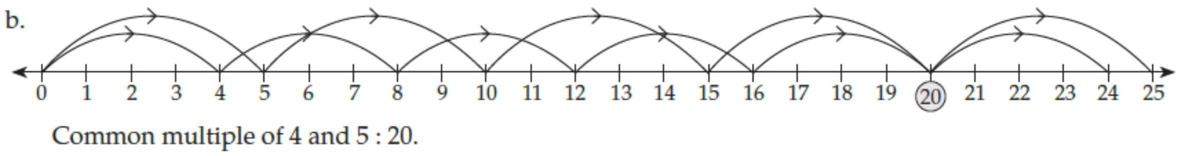
Exercise 5B

4.

a. The first eight multiples of 3 are: 3, 6, 9, 12, 15, 18, 21, and 24.
b. The first eight multiples of 8 are: 8, 16, 24, 32, 40, 48, 56, and 64.
c. The first eight multiples of 11 are 11, 22, 33, 44, 55, 66, 77 and 88.
d. The first eight multiples of 12 are 12, 24, 36, 48, 60, 72, 84 and 96.
e. The first eight multiples of 15 are 15, 30, 45, 60, 75, 90, 105, and 120.
f. The first eight multiples of 20 are 20, 40, 60, 80, 100, 120, 140 and 160.
g. The first eight multiples of 25 are 25, 50, 75, 100, 125, 150, 175 and 200.
h. The first eight multiples of 30 are 30, 60, 90, 120, 150, 180, 210 and 240.

5. The first five even multiples of 9 are: 18, 36, 54, 72, and 90.

6.



The first common multiple of 5 and 4 on the number line is 20. The second will be 40, then 60, 80, 100 and so on.

7.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50

Multiples of 6: 6, 12, 18, 24, 30, 36, 42, 48
Multiples of 8: 8, 16, 24, 32, 40, 48,
Common multiples are 24, 48

8. The odd multiples of 9 that are less than 100 are: 9, 27, 45, 63, 81, and 99.

9. The first 10 multiples of 4: 4, 8, 12, 16, 20, 24, 28, 32, 36, and 40
The first 10 multiples of 6: 6, 12, 18, 24, 30, 36, 42, 48, 54, and 60.
Common multiples of 4 and 6: 12, 24, 36; the smallest is 12.

10. Multiples of 10 from 20 to 150 are: 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, and 150. Thus, there are 14 multiples of 10 from 20 to 150.

11. Multiples of even numbers are always even but, multiples of odd numbers are odd if they are products of odd numbers. So, multiples of odd numbers follow the odd – even pattern. For example: multiples of 2 will always be even -- 2, 4, 6, 8...
However, multiples of 3 will be 3, 6, 9, 12, 15, 18.... following the odd-even pattern.

12. The total number of students in the group is 27.
a. They cannot form equal rows of 2 as 27 being an odd number so after 13 rows one student will be left out.
b. They can form equal rows of 3. 9 such rows will be formed; they can also form 3 rows of nine students each.
c. There is no more way of getting equal rows.

Let's Try! 8

1. 150 has 0 in the ones place, so it is divisible by 2.
    a. Yes
2. 5 is a factor of 5, so it is divisible by 5.
    a. Yes
3. 125 has 5 in the ones place, so it is not divisible by 10.
    b. No

Exercise 5C

3. A number is divisible by 10, if the ones digit is 0.
∴ Numbers between 456 and 522, that are divisible by 10 are:
460, 470, 480, 490, 500, 510, and 520.

4. a. A number is divisible by 2, if the number is even, that is the ones digit is 0, 2, 4, 6, or 8.
∴ The digits that can be filled in the box are: 0, 2, 4, 6, and 8
b. A number is divisible by 5, if the ones digit is 0 or 5.
∴ The digits that can be filled in the box are: 0 and 5.
c. A number is divisible by 10, if the ones digit is 0.
∴ The digits that can be filled in the box is 0.

5. Yes, the roll of cloth measuring 175 m can be cut into equal number of 5m pieces, as 175 is completely divisible by 5.

6. Yes, Rakesh can share 78 marbles equally with his brother as 78 is an even number and is completely divisible by 2. So, it can be shared equally between Rakesh and his brother.

Revision Exercises

Start Up

1.a. The factors of 55 using multiplication are
1 × 55 = 55 , 5 × 11 = 55
Ans: Factors of 55 are 1, 5, 11 and 55.
b. The factors of 90 using multiplication are:
1 × 90, 2 × 45, 3 × 30, 5 × 18, 6 × 15, and 9 × 10
Ans: Factors of 90 are: 1, 2, 3, 5, 6, 9, 10, 15, 18, 30, 45, and 90.
c. Factors of 65 using multiplication are
1× 65, and 5 × 13
Ans: Factors of 65 are: 1, 5, 13 and 65.
d. Factors of 60 using multiplication are
1× 60, 2 × 30, 3 × 20, 4 × 15, 5 × 12, and 6 × 10.
Ans: Factors of 60 are 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, and 60.
e. Factors of 100 using multiplication are:
1 × 100, 2× 50, 4 × 25, 5 × 20, and 10 × 10
Ans: Factors of 100 are: 1, 2, 4, 5, 10, 20, 25, 50 and 100.
f. Factors of 96 using multiplication are
1 × 96, 2× 48, 3 × 32, 4 × 24, 6 × 16, and 8 × 12
Ans: Factors of 96 are: 1, 2, 3, 4, 6, 8, 12, 16, 24, 32, 48, and 96 Among the above 90, 60, and 90 have the greatest number of factors i.e. 12.

2.a. The first eight multiples of 6 are: 6, 12, 18, 24, 30, 36, 42 and 48.

b. The first eight multiples of 7 are: 7, 14, 21, 28, 35, 42, 49 and 56.

c. The first eight multiples of 8 are: 8, 16, 24, 32, 40, 48, 56 and 64.

d. The first eight multiples of 9 are: 9, 18, 27, 36, 45, 54, 63, and 72.

e. The first eight multiples of 10 are: 10, 20, 30, 40, 50, 60, 70 and 80.

f. The first eight multiples of 11 are: 11, 22, 33, 44, 55, 66, 77 and 88.

3.a. 3 is not a factor of 34.

b. 6 is a factor of 66 (6 × 11 = 66).

c. 12 is a factor of 120 (12 × 10 = 120).

d. 11 is not a factor of 111.

e. 7 is a factor of 84 (7 × 12 = 84).

f. 9 is not a factor of 21.

4.a. 18 is not a multiple of 4.

b. 24 is a multiple of 6 (6 × 4 = 24).

c. 72 is a multiple of 8 (8 × 9 = 72).

d. 105 is a multiple of 15 (15 × 7 = 105).

e. 117 is not a multiple of 7.

f. 92 is not a multiple of 5.

Level Up

1.

Divisible by 2	**Divisible by 5**	**Divisible by 10**	**Not divisible by 2, 5 or 10**
6734, 650, 6010, 786, 4418, 2982, 222, 8650, 2118	4565, 650, 6010, 8525, 5555, 8650	650, 6010, 8650	1323, 4989, 7337

2. To find the number of equal groups that the teacher can form, we will have to find out the factors of 36. Using multiplication, the factors of 36 are

1× 36, 2 × 18, 3 × 12, 4 × 9 and 6 × 6

So, factors of 36 are 1, 2, 3, 4, 6, 9, 12, 18 and 36. Thus, at least 7 different groups can be formed.

Note: we will not count 1 and 36 because a group comprises more than one individual and 36 has to be divided into smaller groups. 36 divided by 1 will be 36 itself.

HOTS Questions

1. For the smallest number to be subtracted from 3453 to make it divisible by 2, 5 and 10, the last digit should be 0. So the number will be 3450. Thus 3 has to be subtracted from 3453 to make it divisible by 2, 5 and 10.

2. Since the difference in the number of books and number of shelves on which they are arranged is one, it means that the number 30 has to be divided in such a way that the difference in its factors is one.
   Using multiplication, we find the following factors of 30:
   1×30, 2×15, 3×10, 5×6
   Thus, the number of books in each shelf is 5, and the number of shelves is 6.

Everyday Maths

Beginning the 1st of May, Rohan attended music classes on every third day, so he attended music class on
4th, 7th, 10th, 13th, 16th, 19th, 22nd, 25th, 28th, 31st

Since 1st was the fourth day of the week, the next fourth days of the week would fall on
8th, 15th, 22nd, and 29th.

Thus, we find that Rohan will next attend both classes on 22nd May.